

Solution Requirements

Date	05 July2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The application allows users to access the HDI prediction system through the web interface without complex authentication.	Medium
2	Authorization levels	All users have permission to enter input data and generate HDI predictions.	Low
3	External interfaces	The system interacts with the Flask web application, HTML pages, and the trained HDI.pkl machine learning model.	High

4	Transactions processing	The system accepts user inputs, processes them using the machine learning model, and generates an HDI prediction instantly.	High
5	Reporting	The system displays the predicted HDI score along with its development category (Low, Medium, High, or Very High).	High
6	Business rules, etc.	Prediction is based on Country, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita. Invalid inputs are rejected.	High
7	Compliance to laws or regulations	The application follows basic software development practices and ensures that no personal user information is permanently stored.	Medium
8	Other	The system provides a simple, responsive interface that enables users to perform predictions quickly and accurately.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate predictions quickly after receiving valid inputs.	Prediction time less than 2 seconds
2	Scalability	The application should support multiple users with minimal performance degradation.	Supports multiple concurrent users
3	Security & Data Privacy	User inputs are processed securely without permanent storage of personal information.	No sensitive data stored permanently
4	Reliability & Availability	The application should provide consistent prediction results and remain available during normal operation.	99% application availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy to understand for all users.	User-friendly interface with clear navigation
6	Other	The application should be maintainable, reusable, and easy to update with improved machine learning models.	Modular code structure and documentation

